



Advanced Algorithms 2026

Exercise Session 1

Problems

1 (E-level problem)

Imagine a social network graph $G = (V, E)$ with users in V and an edge between two users denoting a friendship. We say that a set of nodes S is an *influencing set* if the nodes in S (collectively) participate in all the friendships in the network. I want to compute an influencing set of minimum size and I am already given a representative set $R \subseteq E$ of friendships as described in Fig. 1. Prove that V_R is a 2-approximation for the problem of finding an influencing set of minimum size (for G).

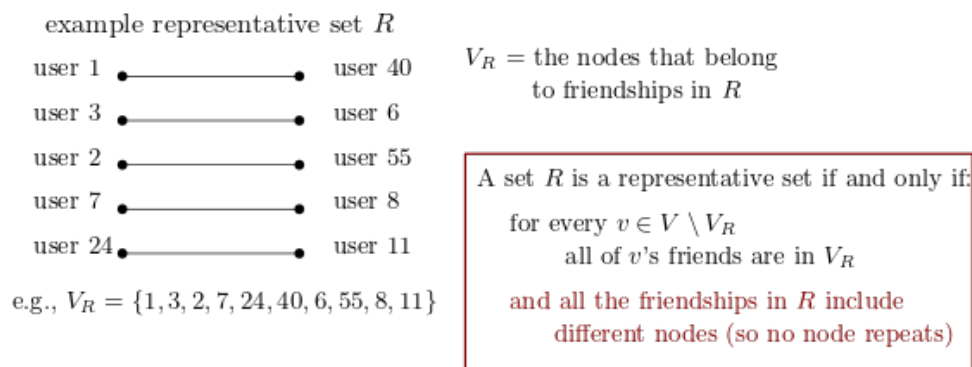


Figure 1: Example and definition of a representative set.

2 (C-level problem)

Per really likes bananas. Today, they were on sale at the supermarket, so he got 100 of them! Now he has a dilemma: how to transport them home in the most efficient way? He knows the weight of each banana and he has (an unlimited supply of) tote bags that can each carry up to 1kg (i.e., you can assume that you can fit any number of bananas in one tote bag as long as their combined weight is at most 1kg). Give a 2-approximation algorithm for using the smallest number of tote bags to carry all of the bananas. You must bound the running time of your algorithm (for any number n of bananas) and prove that it is a 2-approximation.

3 (A-level problem)

Suppose we have a group of n students and m student groups $S_1, \dots, S_m$ that the students belong to. Each student belongs to as many student groups as they like, but no two distinct groups share more than 10 students in common. Give an FPT algorithm for deciding whether there are k student groups that collectively cover the entire student body. Namely, if such a set of student groups exists, then you must output one such set. Otherwise, you must return NO. You must show correctness of your algorithm, bound the running time of your algorithm and show that it is FPT.